

quantum properties such as superposition, measurement disturbance, and the no-cloning theorem.

The no-cloning theorem states that an unknown quantum state cannot be perfectly copied. Consequently, an attacker cannot simply duplicate a quantum state, examine it, and forward an identical copy without potentially introducing detectable changes.

This leads to a powerful security principle: attempting to observe quantum information can disturb it.

How Quantum Key Distribution works

A widely studied QKD protocol is BB84, introduced by Charles Bennett and Gilles Brassard in 1984. In a simplified implementation, a sender, commonly called Alice, encodes information into quantum states of individual photons. The receiver, Bob, measures these states using randomly selected measurement bases.

For example, photon polarization can be used to represent quantum states. Alice may randomly select between two measurement bases and encode a sequence of bits. Bob independently chooses measurement bases when measuring the incoming photons.

After transmission, Alice and Bob communicate over a conventional authenticated classical channel. They publicly compare their selected bases but do not reveal the encoded bit values. Measurements made using compatible bases are retained, while incompatible measurements are discarded. The remaining sequence forms the sifted key.

If an eavesdropper, commonly called Eve, attempts to intercept and measure the photons, she generally does not know which measurement basis was used. Her measurements therefore introduce errors into the transmitted quantum states. Alice and Bob can estimate the resulting error rate and determine whether the communication channel has been compromised.

The role of Quantum Bit Error Rate

An important parameter in QKD is the Quantum Bit Error Rate (QBER). It can be expressed as:

$$QBER = \frac{N_t}{N_e} \times 100\%$$

...where:

- N_e = number of erroneous bits detected
- N_t = Total number bits compared
- QBER = Quantum Bit Error Rate

A low QBER generally indicates that the quantum channel has relatively few errors, whereas a high QBER may indicate significant noise, hardware imperfections, or a potential eavesdropping attempt.

It is important to recognise that QKD does not automatically eliminate every cybersecurity risk. Practical systems must also address device imperfections, authentication, implementation vulnerabilities, detector attacks, and the security of the classical communication channel.

Quantum cryptography beyond QKD

Although QKD receives considerable attention, quantum cryptography is broader than quantum key distribution. Researchers are investigating several quantum-enabled approaches for secure communication, authentication, randomness generation, and secure computation.

Quantum Random Number Generation (QRNG) is another important application. Classical pseudo-random number generators rely on deterministic algorithms, whereas quantum processes can provide sources of fundamentally unpredictable randomness. High-quality random numbers are essential for cryptographic keys, authentication protocols, and security systems.

Quantum technologies are also being investigated for quantum digital signatures, quantum secret

sharing, blind quantum computing, and secure multiparty computation. These developments could eventually create security architectures in which quantum and classical technologies operate together.

Quantum networks: The next frontier

The long-term vision extends beyond point-to-point QKD. Researchers are working towards quantum communication networks, potentially connected through quantum repeaters and quantum memories.

A future quantum network could enable the distribution of entangled quantum states across geographically separated locations. Such networks could form the foundation of a quantum internet, where quantum information is transmitted alongside conventional digital data.

Satellite-based quantum communication is particularly attractive for long-distance links. Terrestrial optical fibres introduce losses that make extremely long-distance quantum communication challenging. Satellites could potentially provide communication paths over much larger distances.

This technology could have significant implications for governments, financial institutions, defence organisations, healthcare networks, data centres, and critical infrastructure.

Challenges on the road to adoption

Despite its promise, quantum cryptography is not a simple replacement for existing encryption. One of the major challenges is hardware scalability. Quantum states are extremely sensitive to environmental disturbances, making reliable transmission and measurement technically demanding.

Other challenges include:

- Photon losses over long communication distances.
- High-performance quantum detectors and sources.
- Integration with existing communication infrastructure.